

Born to Copy, Taught to Trust: What Is Innate in a Language Model

Anonymous ACL submission

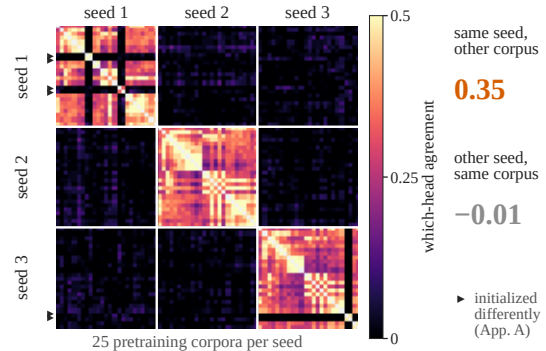
Abstract

Is anything in a language model innate? We answer with a natural experiment hidden in public pretraining suites, where the same random initializations were trained on up to 25 different corpora (DataDecide, 4M–1B parameters) or on two versions of the Pile (Pythia, 70M–12B). What is innate is the model’s anatomy. Which attention heads become induction, previous-token, retrieval and six other kinds of heads is fixed by the random seed before the model reads a word and recurs on every corpus—so faithfully that the head layout alone identifies a model’s seed (98–100%), while the weights keep a correlation of only 0.04 with their initial values. Read blind, the layout even exposes six public checkpoints that were initialized differently from their seed label. Everything the model knows and does is acquired: the corpus sets how strong its circuits are and when they form, no seed is lucky on 11 benchmarks at 14 sizes, and a 1% slice of instruction data teaches a habit—trusting a counterfactual context after the literal template Question: but not after Q:—that shapes context-faithfulness evaluation. As in Lenneberg’s critical period, the innate layout is written in the first 1–2.5% of training, after which not even a switch to a new language, source code, rewrites it; and it recurs across corpora as far as they share content words: what a corpus is about, not how it speaks.

1 Introduction

Whether language is innate or learned has organized the study of acquisition for half a century. One tradition credits children with an innate language faculty (Chomsky, 1965), another builds language from usage with general-purpose learning (Tomasello, 2003), and the window in which a first language is acquired natively has been read as the mark of a maturational schedule (Lenneberg, 1967). Language models have entered this debate as learners whose inductive biases and input can be con-

(a) Innate: the seed chooses the heads



(b) Acquired: 1% instruction data teaches trust after “Question:”

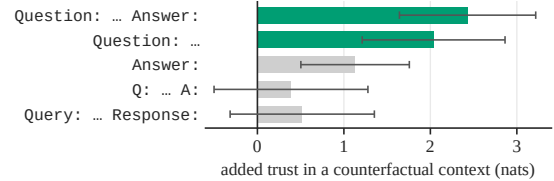


Figure 1: **Innate anatomy, acquired behaviour.** (a) Agreement on *which* head in a layer takes a role, between all 75 one-billion-parameter DataDecide models (3 seeds $\times$ 25 corpora), ordered by seed: models grown from one seed agree on every corpus (bright blocks), models grown on one corpus from different seeds do not. Dark lines (►): six runs initialized differently from their seed label (§3), excluded from the means. (b) 1% of Flan data in pretraining raises trust in a counterfactual context after Question:, not after Q: or Query: (1B, $\pm$ SE).

trolled (Warstadt and Bowman, 2022): they learn possible languages more easily than impossible ones (Kallini et al., 2024), show no critical period for a second language unless one is engineered into them (Constantinescu et al., 2025), and acquire innate-like structural biases from pre-pretraining (Mita et al., 2026).

These studies ask what a language model can learn from its input. We ask what in a language model is innate. A language model has exactly one endowment that precedes its experience—the

random draw of its initial weights, its *seed*—and one experience, its pretraining corpus. We call a property of the trained model *innate* if the seed determines it whatever the corpus, and *acquired* if the corpus determines it whatever the seed. Practice assumes that nothing of the seed survives training: seed variation is averaged away as noise (Reimers and Gurevych, 2017; Dodge et al., 2020; Sellam et al., 2022), training is held to erase the initialization (Das et al., 2026), and interpretable circuits are traced to the data that formed them (Chen et al., 2026). Testing it requires the same seed grown on different corpora, an experiment that public suites have already run. DataDecide (Magnusson et al., 2025) trains three seeds on up to 25 corpora at each of 14 sizes, and Pythia (Biderman et al., 2023) trains a model on the Pile and on the deduplicated Pile from one initialization at every size up to 12B parameters. Their 800 models, with controlled pre-training runs of our own, let us separate the two.

What is innate is the anatomy (Figure 1a). Which heads become a model’s induction, previous-token and retrieval heads—the copying circuitry behind in-context learning (Olsson et al., 2022)—is fixed by its seed and recurs on every corpus, for all nine head roles we measure, at every size and in both suites. The layout identifies the seed with 98–100% accuracy although the trained weights keep a correlation of only 0.04 with their initial values; read blind, it even exposes six DataDecide checkpoints that were initialized differently from their seed label. The weights forget the seed; the circuits remember it. Language models are *born to copy*: where they will copy is decided before they read a word.

What is acquired is everything the model knows and does. How strong its circuits are, when they form, what it represents and how it performs on 11 benchmarks come from the corpus, with no main effect of the seed. One acquired habit shows how specific this teaching is (Figure 1b): a 1% slice of instruction data teaches a model to trust a counterfactual context when the question is introduced by *Question:*, not by the equivalent *Q:*, across sizes, datasets and two independent training stacks. Language models are *taught to trust*, and how much they trust a context depends on the template a benchmark happens to use.

The two meet in a critical period. As in Lenneberg’s account, the innate layout is written early: in controlled pretraining, switching the corpus or adding noise during the first 1% of training

re-draws it, whereas from 2.5% on even noise as large as the weights returns every role to its head. The layout then recurs across corpora as far as they share *content words*—what the corpora are about—while their function words, how they speak, play no part.

Our contributions are:

- **A natural experiment on nature and nurture in language models**, built from seed $\times$ corpus crossings in public suites at 21 model sizes from 4M to 12B parameters, with a measurement that separates which layer from which head hosts a role and a symmetry argument under which the corpus provably cannot choose the head (§2).
- **Innate anatomy, acquired function**: head placement is inherited from the seed, identifies it and exposes checkpoints whose initialization differs from their label; knowledge, timing, benchmarks and behaviour are learned, and no seed is lucky (§3, §4).
- **A template habit learned from 1% of pretraining data**, keyed to *Question:*, present in public base models, with consequences for context-faithfulness evaluation (§4.1).
- **A critical period shaped by what the corpus is about**: the layout is written in the first 1–2.5% of training and shared across corpora according to their content words, not their function words (§5, §6).

2 A Natural Experiment

Seeds crossed with corpora. DataDecide (Magnusson et al., 2025) pretrains OLMo-style models at 14 sizes (4M–1B parameters) on 25 corpus recipes—C4, DCLM and its quality-filtered variants, Dolma 1.6/1.7 and their ablations, Falcon RefinedWeb variants, FineWeb-Edu/-Pro and mixtures—each with three seeds. Within a size, a seed’s initial weights are shared across recipes, so each seed has been grown on up to 25 corpora; we confirm every run’s initialization from its released weights (Appendix A). Pythia (Biderman et al., 2023) trains a standard and a deduplicated model from identical initial weights at every size; we use 70M–12B, with the independently initialized PolyPythias (van der Wal et al., 2025) as a different-seed reference.

Pairs. A crossing contains three kinds of model pairs: *same seed, other corpus*; *other seed, same corpus*; and both different. A property is **innate**

if same-seed pairs agree more than same-corpus pairs, **acquired** if the reverse holds, and universal if all pairs agree.

Reading the anatomy. For each model we compute a [layer $\times$ head] map for each of nine head roles from fixed probe material (Appendix B): induction, previous-token, retrieval (attention from the answer position to the context entity in knowledge-conflict prompts), attention sink (Xiao et al., 2024), duplicate-token, current-token, two-back and delimiter heads, and, as a measure that involves no attention pattern, each head’s OV copying score computed from its weights alone (Elhage et al., 2021). A role map answers two questions at once: which *layer* hosts the role and which *head* within that layer takes it. We measure the first with the rank correlation of per-layer profiles and the second, the *which-head agreement*, with the Spearman correlation of the head scores within each layer, averaged over layers.

The corpus cannot choose the head. Heads within a layer are interchangeable: permuting them, with their query, key, value and output weights, leaves the network’s function unchanged, the initialization distribution is invariant to the permutation, and gradient-based optimizers respect it. Whatever the corpus, each head is therefore equally likely to become, say, the induction head, and two independently initialized models agree on which head does only at chance—a which-head agreement of exactly zero in expectation (Lemma 1, Appendix C). A corpus can never *choose* a head; agreement above zero between models trained on different corpora is carried by their shared seed. The empirical question is whether the seed’s choice survives a change of corpus.

Reading behaviour. For function we use DataDecide’s 11 benchmark evaluations (Table 8), circuit strength and developmental timing, and *context trust*: on knowledge-conflict prompts that state a counterfactual fact, the log-probability margin of the counterfactual over the memorized answer, on facts the model knows (Li et al., 2025; Du et al., 2024).

3 Innate: Where the Circuits Live

Finding 1: The seed decides which head takes each role. In the 1B crossing, two models grown from the same seed on different corpora agree on which head within a layer takes each role, whereas

(a) DataDecide 1B, 3 seeds $\times$ 25 corpora (69 models)

Role	same seed	other seed	gap, 95% CI
Two-back	0.43	0.00	[0.40, 0.46]
Previous-token	0.42	−0.01	[0.40, 0.47]
Duplicate-token	0.36	−0.02	[0.36, 0.41]
Induction	0.33	−0.02	[0.32, 0.37]
Current-token	0.32	−0.02	[0.31, 0.36]
Retrieval	0.31	0.00	[0.28, 0.35]
Delimiter	0.22	−0.01	[0.20, 0.25]
Attention sink	0.18	−0.01	[0.16, 0.23]
OV copying (weights)	0.25	−0.01	[0.25, 0.29]

(b) Pythia, one seed: Pile vs. deduplicated Pile

Size	induction	prev.-token	retrieval	null
70M	−	0.60	0.58	0.26
160M	0.49	0.58	0.53	0.14
410M	0.47	0.55	0.45	0.09
1B	0.56	0.72	0.58	0.16
1.4B	0.48	0.54	0.50	0.09
6.9B	0.53	0.60	0.49	0.06
12B	0.38	0.51	0.38	0.04
12B, at 2.1%	0.55	0.60	0.45	0.04

Table 1: Head placement is inherited from the seed. Which-head agreement (within-layer Spearman of head scores). (a) All nine roles at 1B for models grown from the same seed on different corpora and from different seeds on the same corpus; the gap’s interval is a recipe bootstrap. (b) Pythia’s two models from one initialization; null: 95th percentile after permuting heads within layers; no induction head at 70M.

two models grown on the same corpus from different seeds agree at chance (Table 1a, Figure 1a). Concretely, under one seed a single head of layer 7 is the strongest induction head on 14 of the 25 corpora, where chance predicts 1.6. The pattern holds for all nine roles (0.18–0.43 against −0.02 to 0.00), including the copying score computed from the weights alone, with five seeds at 1B, and on corpus pairs that share no data source. Decomposing every head’s rank into seed, corpus and residual components gives a median corpus component of exactly zero: the seed effect is significant for 66–79% of heads, the corpus effect for 4–7%, the false-positive rate. Inheritance holds at all 14 DataDecide sizes, growing with size (Figure 2a; Spearman with log parameters 0.81–0.92), and in the second suite: Pythia’s two models from one seed agree at every size from 70M to 12B, far above a permutation null (Table 1b), while independently initialized PolyPythias seeds agree at chance.

Finding 2: The layout is a birth certificate. We hold out a model, compare its head layout with the models grown from each seed on the *other* corpora, and assign it to the seed whose models it resem-

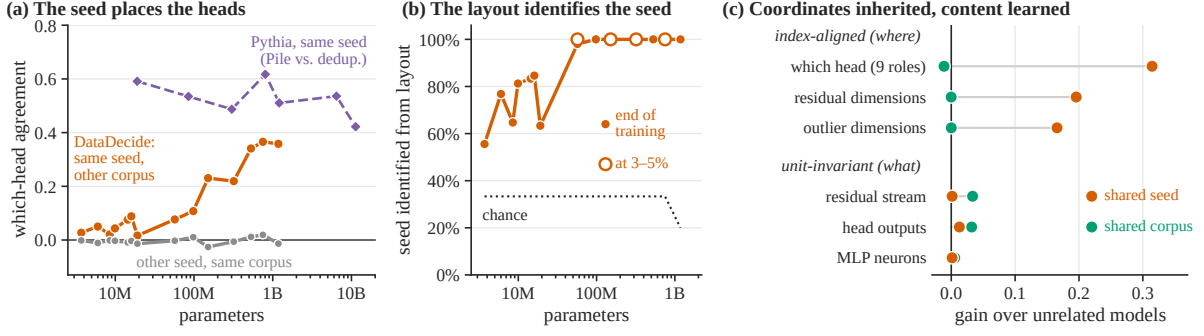


Figure 2: **The seed places the heads.** (a) Which-head agreement (mean over induction, previous-token and retrieval maps) for DataDecide model pairs grown from the same seed on different corpora and from different seeds on the same corpus, at 14 sizes, and for Pythia models grown from one seed on the Pile and on the deduplicated Pile, up to 12B. (b) Identifying a model’s seed from its head layout, leaving its corpus out: accuracy at the end of training and at 3–5% of training. (c) Similarity gained over unrelated models by sharing the seed or the corpus: the seed is shared in index-aligned quantities (which head, which residual dimension), the corpus in unit-invariant ones (CKA, best match).

bles most. From 60M parameters on, this identifies the seed with 98–100% accuracy (chance: 33%, or 20% with five seeds), including all 69 one-billion-parameter models, and already at 3–5% of training (Figure 2b, Table 5); in Pythia the true sibling ranks first among ten candidates at all three sizes with independently initialized alternatives. Applied without labels to the 75 one-billion-parameter models, the layout singles out six runs that agree with none of their nominal siblings but pairwise with each other (Figure 1a, dark lines). Their published weights confirm it: the six runs were not initialized from their labelled seed’s weights (correlation 0.000), and each pair shares an initialization of its own (Table 4). SeedPrints (Tong et al., 2026) identifies a lineage of weights from output biases; the anatomy identifies independently trained siblings whose weights have long diverged.

Finding 3: The weights forget the seed; the coordinates do not. Final weights correlate with their initial values at only 0.03–0.09, and same-seed models trained on different corpora are not linearly connected: the loss barrier between them (Frankle et al., 2020) is 7.5 nats, close to the 7.8 between unrelated models and above the 7.0 between models that share only their corpus. Their representations are no more alike than unrelated models’ under unit-invariant measures—CKA (Kornblith et al., 2019) and best-match similarity, both highest for models that share a corpus—and MLP neurons do not inherit their identity (0.008). What the seed fixes is a coordinate system: which head hosts which role, which residual dimensions carry

Property	Set by	Key statistic
Which head takes a role	seed	0.18–0.43 vs. 0.00
Residual-stream coordinates	seed	0.20 vs. 0.00
Layer that hosts a role	universal	0.4–0.7, all pairs
Copying algorithm	universal	75 / 75 models
Circuit strength	corpus	4 / 6 measures
Onset of induction heads	corpus	81% vs. 0%
Representation content	corpus	CKA, best match
11 benchmarks, 14 sizes	corpus	seed sig. 4–8%
Knowledge-conflict behaviour	corpus	seed sig. 0 / 12
Question: habit	corpus	+2.4 nats
Weight values	neither	$r = 0.04$

Table 2: **What is innate and what is acquired.** For each property, the factor that determines it and the key statistic: same-seed vs. other-seed agreement; variance share of corpus vs. seed; or share of cells with a significant seed effect (5% is the false-positive rate).

variance (0.20 vs. 0.00) and which carry the massive outlier activations (0.17 vs. 0.00; Figure 2c).

What is universal. The layer that hosts each role is shared by all models regardless of seed or corpus (layer-profile agreement 0.4–0.7 for both pair types), and every 1B model implements in-context copying with the same algorithm, a previous-token head composing with an induction head (75 of 75 models). What the seed decides is exactly *which* of the interchangeable heads in a layer takes each role.

4 Acquired: What the Model Knows and Does

Finding 4: Everything functional comes from the corpus. How strongly a circuit is expressed

Ending after the context	count in Dolma 1.7	Flan effect	SE
Question: <i>q</i> Answer:	11.3M / 9.6M	+2.43	0.79
Question: <i>q</i>	11.3M	+2.04	0.83
Answer:	9.6M	+1.13	0.63
Q: <i>q</i> A:	38.4M / 6.9M	+0.39	0.89
Query: <i>q</i> Response:	0.15M / 0.75M	+0.52	0.83
Question: <i>q'</i> Answer:	as first row	+3.06	1.42
declarative (no question)	–	+0.20	0.50

Table 3: **The habit is keyed to Flan’s literal template.** Flan minus no-Flan models (1B, three seeds each): added trust in the counterfactual context relative to a declarative ending, in nats. Example: *The capital of France is Rome.* Question: *What is the capital of France?* Answer: *The capital of France is ...*; *q'* asks about another entity. Marker counts: occurrences in the pretraining corpus. The last row is the raw trust difference for the declarative ending.

is set by the corpus (significant corpus effect on 4 of 6 strength measures, seed component ≈ 0). So is when it forms: in controlled pretraining with six seeds and three corpora, the corpus explains 81% of the variance in the onset of induction heads and the seed none ($p = 0.0001$ vs. 0.47), whereas previous-token heads form at the same time under every seed and corpus. Across 11 benchmarks $\times$ 14 sizes, the seed has a significant main effect in 3.9% (accuracy) and 7.8% (per-byte likelihood) of the 154 cells—the false-positive rate—and the corpus in 75% and 98% (median variance components: seed 0.000, corpus 0.49–0.83; Table 8). Behaviour under knowledge conflict shows no seed effect either (0 of 12 conditions). There are no lucky seeds: all seed variance in function is seed $\times$ corpus interaction, so a seed that does well on one corpus is no more likely to do well on another (Table 2).

4.1 A Template Habit from 1% of the Data

The clearest window on what a corpus teaches is a component that some corpora contain and others do not. Dolma 1.7 contains about 1–2% of Flan instruction data (Longpre et al., 2023; Soldaini et al., 2024), and DataDecide includes a Dolma 1.7 recipe with Flan removed, trained from the same seeds. We compare the two on knowledge-conflict prompts from ParaConflict (Li et al., 2025), in which one sentence asserts a counterfactual fact (“The capital of France is Rome.”) and is followed either by a declarative continuation or by a question in Flan’s format (Question: What is the capital of France? Answer:).

Finding 5: 1% of instruction data teaches the model to trust the context after Question:. Removing Flan lowers the 1B models’ trust in the counterfactual context by 1.3–3.5 nats across four kinds of context whenever the prompt is a question, and leaves declarative prompts and the models’ knowledge of the facts untouched. The effect follows the question format (+2.23 nats), not how often the context states the counterfactual, the argument around it or its length. In behavioural terms, the share of questions on which a model ignores a one-sentence context and answers from memory falls from 14.3% to 1.9%.

The habit is keyed to a literal template. The trigger is Flan’s own wording (Table 3). Question: alone carries most of the effect, whereas the equivalent Q: ... A: and Query: ... Response: leave trust where it is—although Q: occurs three times as often as Question: in Dolma 1.7. The habit does not read the question: a question about a different entity triggers it as strongly.

It holds across sizes, datasets and stacks. The habit is present at six of seven sizes from 60M to 1B (Figure 3a), appears by 3.6% of pretraining and grows to +2.4 nats (Figure 3b), and replicates on PopQA counterfactuals (+1.91; Mallen et al., 2023) and on NQ-Swap (Longpre et al., 2021), where on natural reading-comprehension passages it is strongest after Question: and extends to Q: (Figure 3d). In an independent stack, OLMo 2 (Team OLMo et al., 2024), mid-training on a mixture containing FLAN raises it from ≈ 0 to +1.4–2.3 nats in each of three independent runs, while trust under declarative prompts falls (Figure 3c). Taught early, the habit takes a small dose; taught late, a large one: the from-scratch 1B models have it after about 50M Flan tokens, continued pretraining of a Flan-free 60M model on 90M Flan tokens does not install it ($+0.04 \pm 0.04$), and OLMo 2 acquires it within its first 5B tokens of mid-training, which contain about 0.8B FLAN tokens. Unlike the roles of §3, the heads that implement it are consistent within a model (split-half 0.93) but not placed by the seed (0.11 vs. 0.02): a function taught by specific data is hosted flexibly.

Consequences for evaluation. Context-faithfulness and knowledge-conflict benchmarks commonly pose their questions in exactly this format, and every public base model we tested trusts a counterfactual context more after Question:

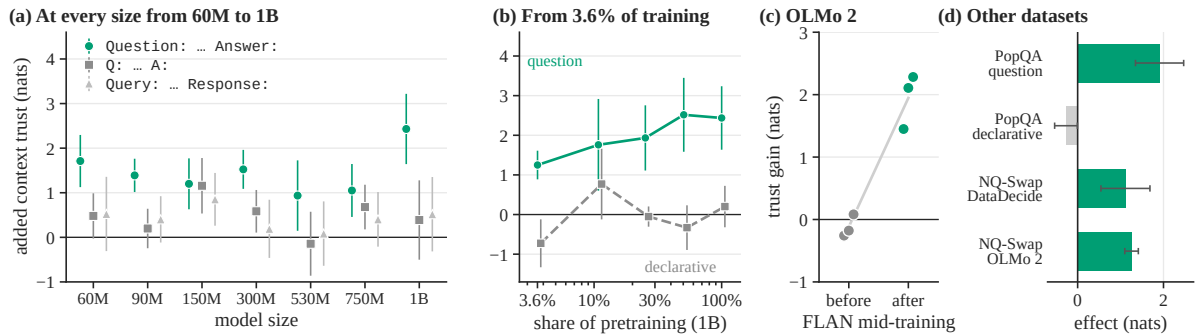


Figure 3: **The Question: habit is learned from pretraining data.** Added trust in a counterfactual context (log-probability margin of the counterfactual over the memorized answer, relative to a declarative prompt; nats, $\pm$ SE). (a) Flan minus no-Flan DataDecide models at every size, for the Flan template and two equivalent templates. (b) During pretraining of the 1B models: present at the first checkpoint (3.6%); declarative prompts unaffected. (c) OLMo 2 1B before and after mid-training on a mixture with 16.6% FLAN (three mid-training runs). (d) PopQA counterfactuals and NQ-Swap (Question: vs. Q:).

than in a declarative prompt, by 1.0–4.6 nats, most of all the one pretrained with Flan (Table 10). A model’s score on such benchmarks therefore reflects, besides its faithfulness, a habit that a small slice of its pretraining data installed and that a change of template removes. This gives prompt-template sensitivity (Sclar et al., 2024; Hua et al., 2025; Liu and Chu, 2026) a concrete origin in the pretraining corpus, beside the data properties that shape knowledge use (Kim et al., 2026).

5 A Critical Period for the Anatomy

Lenneberg’s critical period describes a window in which language is acquired natively and after which the native language is no longer overwritten. Constantinescu et al. (2025) found that language models show no such window for the *behaviour* of a second language. Their anatomy has one.

Finding 6: The layout is written in the first 1–2.5% of training. We pretrain small Llama-style models on C4 and branch them at step k of 10,000 by switching the corpus to source code or by adding Gaussian noise to every weight (Figure 4a; Table 6). Up to $k = 100$ (1%), either intervention re-draws the layout: after a switch to code at 1%, agreement with the untouched run is 0.09, and noise as large as the weights leaves 0.03 at step 0 and 0.36 at 1%. From $k = 250$ (2.5%) on, the layout is an attractor: noise as large as the weights themselves returns the roles to their heads (0.85 at 2.5%, 0.95–0.98 from 5%, the level of an exact rerun), and a switch to code preserves most of it (0.60

at 2.5%, rising to 0.95 at 80%). A 12-layer model shows the same window (a code switch at 1% keeps 0.40, at 10% 0.84; trained on code from scratch, ≈ 0). The public suites lock in during the same window (Figure 4b): the same-seed agreement of the DataDecide 1B models reaches its final level by 3.6% of training, Pythia’s previous-token heads by about 2%, and Pythia-12B’s two models already share their layout at 2.1% (Table 1b). Before training, the layout cannot be read off the weights—no statistic of a head’s initial weights or initial gradient predicts which head will win—so it is written collectively, by the first percent of experience.

Why larger models are more innate. Which head wins is a symmetry-breaking event, decided by the initial condition and by the noise present at the moment of breaking. The lower the stochastic-gradient temperature—the learning rate per token of batch (Smith and Le, 2018; Jastrzębski et al., 2018)—the more the seed alone decides: at fixed size, models that differ only in the order of their batches agree 0.43 with 16-sequence batches and 0.90 with 512-sequence batches, and same-seed agreement across corpora rises from 0.01 to 0.22 (Appendix F). Standard scaling recipes train larger models colder (DataDecide’s learning rate per batch token falls 140-fold from 4M to 1B), and a 30 $\times$ larger model at fixed temperature inherits no more; Pythia, trained with large batches at every size, is innate already at 70M.

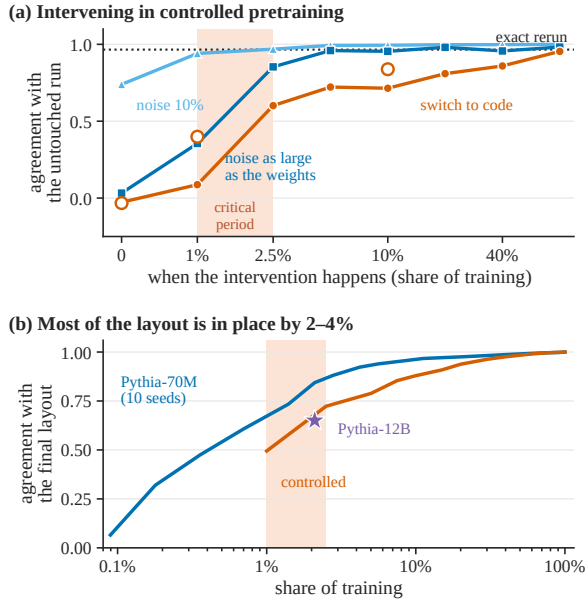


Figure 4: **A critical period.** (a) Controlled pretraining (4 layers $\times$ 8 heads, 10k steps): agreement of the final layout with the untouched run after switching the corpus to code or adding Gaussian noise to every weight at a given step; hollow circles: a 12-layer model. (b) Agreement of the layout with its own final layout during training in public suites and controlled runs; star: Pythia-12B at 2.1%. Shaded: 1–2.5% of training.

6 What the Corpus Is About, Not How It Speaks

If the layout is chosen while the first percent of the corpus is read, two corpora should yield the same layout to the extent that they present the same early statistics.

Finding 7: The closer two corpora’s content words, the more faithfully the layout recurs. Across DataDecide’s corpus pairs, the Jensen–Shannon distance between their unigram distributions predicts how faithfully a seed’s layout is reproduced on both. The law holds at every size from 60M on and sharpens with scale (Spearman -0.20 to -0.33 at 60M, -0.65 to -0.83 from 300M on, -0.76 to -0.80 across the 300 corpus pairs at 1B; Mantel $p < 0.001$ from 90M; Table 5), on pairs that share no data source (-0.36 to -0.82), and after partialling out source overlap. Bigram distance predicts less (-0.55 to -0.62 at 1B): the signal lies in which words a corpus uses. Splitting the vocabulary into function words, the statistical scaffold of grammar (Yang et al., 2026), and content words separates how a corpus speaks from what it is about. The law is carried by content

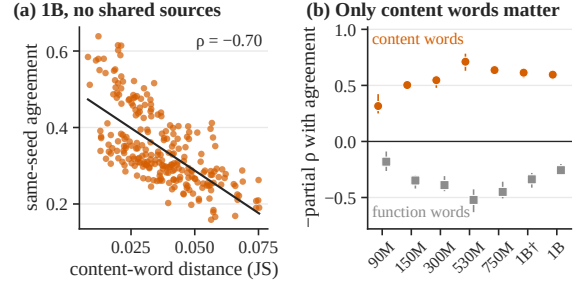


Figure 5: **The layout is shared as far as content is shared.** (a) 1B pairs of DataDecide corpora that share no data source: Jensen–Shannon distance between their content-word distributions vs. the which-head agreement of same-seed models trained on them. (b) Negative partial Spearman correlation between distance and agreement for content words (controlling for function words) and for function words (controlling for content words), by size; bars span the three roles; †: 1B with five seeds at 10% of training.

words (Figure 5): content-word distance predicts agreement at every size ($\rho -0.23$ to -0.69), also when function-word distance is controlled (partial $\rho -0.25$ to -0.78), whereas function-word distance, once content words are controlled, no longer predicts lower agreement at any size (partial $\rho \geq 0.09$)—English web corpora all use their function words alike. Beyond this range the law holds in the extreme: in controlled pretraining, C4 and scientific papers share the layout (0.19 ± 0.06 over nine seeds), whereas any pair involving source code re-draws it (≈ 0), and Pythia’s two versions of the Pile, nearly identical in content, share it most of all.

7 Related Work

Language models as learners. Language models are increasingly used to test claims about acquisition: what is learnable from input (Warstadt and Bowman, 2022; Kallini et al., 2024), the order in which words and structures are acquired (Chang and Bergen, 2022; Chen et al., 2024), critical periods (Constantinescu et al., 2025; Mita et al., 2025) and innate constraints (Mita et al., 2026). We study the learner itself and find an innate component that is anatomical rather than linguistic, together with a critical period that the behaviour does not show.

Seeds in NLP. Seed variation is a known source of irreproducibility in fine-tuning and pretraining (Reimers and Gurevych, 2017; Dodge et al., 2020; McCoy et al., 2020; Sellam et al., 2022), studied as

noise to be modelled (Benito-Santos et al., 2025) or as convergence across training (Fehlauer et al., 2025; van der Wal et al., 2025). SeedPrints (Tong et al., 2026) shows that a seed leaves an identifiable output signature. Varying only the seed, these works cannot tell what the seed contributes beyond noise; crossing seeds with corpora shows that it contributes a heritable anatomy and no function.

Universality and stability of circuits. Algorithms recur across models while components vary (Chughtai et al., 2023; Gurnee et al., 2024; Tigges et al., 2024), attention heads differ across retraining (Bali et al., 2026) and specialize along developmental trajectories (Wang et al., 2025), and head functions can be read from parameters (Elhelo and Geva, 2025). Components vary because they follow the seed, and recur wherever seed and corpus content are shared.

From data to mechanisms. Data properties shape whether in-context learning emerges (Chan et al., 2022; Singh et al., 2024) and which training examples form a head (Chen et al., 2026). These effects concern whether and when a mechanism forms; where it forms is the seed’s choice.

Knowledge conflicts and prompt format. Whether a model follows a counterfactual context or its memory depends on the context’s form and the model’s confidence (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024), is mediated by competing mechanisms (Ortu et al., 2024), and shifts with the prompt’s wording (Zhou et al., 2023; Sclar et al., 2024; Hua et al., 2025; Liu and Chu, 2026); synthetic pretraining shows which data properties make models use both sources (Kim et al., 2026). We trace one such format effect to a 1% slice of natural pretraining data and to a literal template.

8 Implications and Conclusion

For evaluation and reproducibility. No seed is lucky across corpora, so seed selection cannot be transferred between recipes, and sharing seeds across recipes does not reduce the noise of recipe comparisons; mixed-effects analyses should model seed $\times$ corpus interaction rather than a seed main effect. Context-faithfulness benchmarks should report their results across question templates.

For interpretability and data attribution. A circuit diagram of one model is a statement about one seed: its algorithm and layers generalize, its

head labels do not, except to siblings grown from the same seed on similar corpora. Ablating the ten most important heads of a same-seed sibling trained on another corpus does 19.5% of the damage of ablating the model’s own; heads found in a different-seed model trained on the *same* corpus do 2.0%. Data attribution explains whether a circuit forms, not where it lives, and comparisons of models trained on different data should match their seeds. Seed-matched siblings are, in turn, a free control: grown from one seed on different corpora, they share an anatomy, so every difference between their circuits is the data’s doing.

For model provenance. Because the anatomy outlives the weights, the layout certifies whether two released models grew from one initialization, with no access to their training; read this way, it found the six DataDecide runs whose initialization differs from their label.

For training. The two levers that set how much of a model is innate are in the practitioner’s hands: colder training, with larger batches or smaller learning rates, lets the seed fix the layout and makes circuits reproducible across corpora, whereas changing the corpus within the first percent of training draws a new one.

For the nature–nurture question. In the terms of the debate we began with, a language model’s innate endowment is not a grammar but an anatomy: a layout of where its computations will live, drawn at random, written into the network during a brief critical period, and indifferent to the language that fills it. Everything linguistic—what the model knows, how it behaves, which templates it trusts—is acquired, and acquired with surprising literalness. The classical picture of the child is reversed: there the knowledge is innate and its placement flexible; here the placement is innate and the knowledge learned.

Conclusion. A language model is born with an anatomy and taught everything else. Its seed decides, in the first percent of training, which of many interchangeable heads will carry each of its copying circuits, and this decision survives the change of corpus, the forgetting of the initial weights and the rest of training. Its corpus decides what those circuits do—down to a habit of trusting whatever follows Question:.

Limitations

We study decoder-only transformers pretrained on English corpora, up to 12B parameters. The natural experiment extends directly to multilingual suites, where function words differ across languages, and to other architectures as seed-crossed suites become available for them. Our interventions on the critical period use models of up to 12 layers; intervening on larger models during their first percent of training is the natural next step. The template habit opens two questions we leave to future work: why models key on Question: rather than an equivalent marker, and which other instruction-data families install habits of their own. What at initialization decides which head will win is likewise open.

Ethical Considerations

We analyse publicly released models and datasets under their licences and train small models on public corpora; no human subjects are involved. We report our audit of public checkpoints to support their correct use.

References

- Karan Bali, Jack Stanley, Praneet Suresh, and Danilo Bzdok. 2026. [Quantifying LLM attention-head stability: Implications for circuit universality](#). arXiv preprint arXiv:2602.16740.
- Alejandro Benito-Santos, Adrian Ghajari, and Víctor Fresno. 2025. [Robust estimation of population-level effects in repeated-measures NLP experimental designs](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 33076–33089, Vienna, Austria. Association for Computational Linguistics.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. [Pythia: A suite for analyzing large language models across training and scaling](#). In *Proceedings of the 40th International Conference on Machine Learning*.
- Stephanie C. Y. Chan, Adam Santoro, Andrew K. Lampinen, Jane X. Wang, Aaditya Singh, Pierre H. Richemond, Jay McClelland, and Felix Hill. 2022. [Data distributional properties drive emergent in-context learning in transformers](#). In *Advances in Neural Information Processing Systems*.
- Tyler A. Chang and Benjamin K. Bergen. 2022. [Word acquisition in neural language models](#). *Transactions of the Association for Computational Linguistics*, 10:1–16.
- Angelica Chen, Ravid Shwartz-Ziv, Kyunghyun Cho, Matthew L. Leavitt, and Naomi Saphra. 2024. [Sudden drops in the loss: Syntax acquisition, phase transitions, and simplicity bias in MLMs](#). In *International Conference on Learning Representations*.
- Jianhui Chen, Yuzhang Luo, and Liangming Pan. 2026. [Mechanistic data attribution: Tracing the training origins of interpretable LLM units](#). In *Proceedings of the 43rd International Conference on Machine Learning*.
- Noam Chomsky. 1965. *Aspects of the Theory of Syntax*. MIT Press, Cambridge, MA.
- Bilal Chughtai, Lawrence Chan, and Neel Nanda. 2023. [A toy model of universality: Reverse engineering how networks learn group operations](#). In *Proceedings of the 40th International Conference on Machine Learning*.
- Ionut Constantinescu, Tiago Pimentel, Ryan Cotterell, and Alex Warstadt. 2025. [Investigating critical period effects in language acquisition through neural language models](#). *Transactions of the Association for Computational Linguistics*, 13:96–120.
- Mohua Das, Pierfrancesco Beneventano, Shibshankar Dey, Gareth H. McKinney, and Tomaso Poggio. 2026. [Do deep networks forget initialization? a forgetting-time view of practical inductive bias](#). arXiv preprint arXiv:2605.29152.
- Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. 2020. [Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping](#). arXiv preprint arXiv:2002.06305.
- Kevin Du, Vésteinn Snæbjarnarson, Niklas Stoehr, Jennifer White, Aaron Schein, and Ryan Cotterell. 2024. [Context versus prior knowledge in language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13211–13235, Bangkok, Thailand. Association for Computational Linguistics.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, and 6 others. 2021. [A mathematical framework for transformer circuits](#). Transformer Circuits Thread.
- Amit Elhelo and Mor Geva. 2025. [Inferring functionality of attention heads from their parameters](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 17701–17733, Vienna, Austria. Association for Computational Linguistics.
- Finlay Fehlaue, Kyle Mahowald, and Tiago Pimentel. 2025. [Convergence and divergence of language models under different random seeds](#). In *Proceedings*

678	of the 2025 Conference on Empirical Methods in	Linguistics (Volume 1: Long Papers), pages 44362–	733
679	Natural Language Processing, pages 32982–32991,	44388, San Diego, California, United States. Associ-	734
680	Suzhou, China. Association for Computational Lin-	ation for Computational Linguistics.	735
681	linguistics.		
682	Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M.	Shayne Longpre, Le Hou, Tu Vu, Albert Webson,	736
683	Roy, and Michael Carbin. 2020. Linear mode connec-	Hyung Won Chung, Yi Tay, Denny Zhou, Quoc V. Le,	737
684	tivity and the lottery ticket hypothesis . In <i>Proceed-</i>	Barret Zoph, Jason Wei, and Adam Roberts. 2023.	738
685	<i>ings of the 37th International Conference on Machine</i>	The Flan collection: Designing data and methods for	739
686	<i>Learning</i> .	effective instruction tuning . In <i>Proceedings of the</i>	740
		<i>40th International Conference on Machine Learning</i> .	741
687	Wes Gurnee, Theo Horsley, Zifan Carl Guo, Tara Rezaei	Shayne Longpre, Kartik Perisetla, Anthony Chen,	742
688	Kheirkhah, Qinyi Sun, Will Hathaway, Neel Nanda,	Nikhil Ramesh, Chris DuBois, and Sameer Singh.	743
689	and Dimitris Bertsimas. 2024. Universal neurons in	2021. Entity-based knowledge conflicts in question	744
690	GPT2 language models . <i>Transactions on Machine</i>	answering . In <i>Proceedings of the 2021 Conference</i>	745
691	<i>Learning Research</i> .	<i>on Empirical Methods in Natural Language Process-</i>	746
		<i>ing</i> , pages 7052–7063, Online and Punta Cana, Do-	747
692	Andong Hua, Kenan Tang, Chenhe Gu, Jindong Gu,	minican Republic. Association for Computational	748
693	Eric Wong, and Yao Qin. 2025. Flaw or artifact?	Linguistics.	749
694	rethinking prompt sensitivity in evaluating LLMs .		
695	In <i>Proceedings of the 2025 Conference on Empiri-</i>	Ian Magnusson, Nguyen Tai, Ben Bogin, David Heine-	750
696	<i>cal Methods in Natural Language Processing</i> , pages	man, Jena D. Hwang, Luca Soldaini, Akshita Bha-	751
697	19889–19899, Suzhou, China. Association for Com-	gia, Jiacheng Liu, Dirk Groeneveld, Oyvind Tafjord,	752
698	putational Linguistics.	Noah A. Smith, Pang Wei Koh, and Jesse Dodge.	753
		2025. DataDecide: How to predict best pretraining	754
699	Stanisław Jastrzębski, Zachary Kenton, Devansh Arpit,	data with small experiments . In <i>Proceedings of the</i>	755
700	Nicolas Ballas, Asja Fischer, Yoshua Bengio, and	<i>42nd International Conference on Machine Learning</i> .	756
701	Amos Storkey. 2018. Three factors influencing min-		
702	ima in SGD . In <i>Artificial Neural Networks and Ma-</i>	Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das,	757
703	<i>chine Learning – ICANN 2018</i> .	Daniel Khachabi, and Hannaneh Hajishirzi. 2023.	758
704	Julie Kallini, Isabel Papadimitriou, Richard Futrell,	When not to trust language models: Investigating	759
705	Kyle Mahowald, and Christopher Potts. 2024. Mis-	effectiveness of parametric and non-parametric mem-	760
706	sion: Impossible language models . In <i>Proceedings</i>	ories . In <i>Proceedings of the 61st Annual Meeting of</i>	761
707	<i>of the 62nd Annual Meeting of the Association for</i>	<i>the Association for Computational Linguistics (Vol-</i>	762
708	<i>Computational Linguistics (Volume 1: Long Papers)</i> ,	<i>ume 1: Long Papers)</i> , pages 9802–9822, Toronto,	763
709	pages 14691–14714, Bangkok, Thailand. Association	Canada. Association for Computational Linguistics.	764
710	for Computational Linguistics.		
711	Minsung Kim, Dong-Kyum Kim, Jea Kwon, Nakyeong	Nathan Mantel. 1967. The detection of disease cluster-	765
712	Yang, Kyomin Jung, and Meeyoung Cha. 2026. How	ing and a generalized regression approach. <i>Cancer</i>	766
713	training data shapes the use of parametric and in-	<i>Research</i> , 27(2):209–220.	767
714	context knowledge in language models . In <i>Proceed-</i>		
715	<i>ings of the 64th Annual Meeting of the Association for</i>	R. Thomas McCoy, Junghyun Min, and Tal Linzen.	768
716	<i>Computational Linguistics (Volume 1: Long Papers)</i> ,	2020. BERTs of a feather do not generalize together:	769
717	pages 23242–23257, San Diego, California, United	Large variability in generalization across models with	770
718	States. Association for Computational Linguistics.	similar test set performance . In <i>Proceedings of the</i>	771
		<i>Third BlackboxNLP Workshop on Analyzing and In-</i>	772
719	Simon Kornblith, Mohammad Norouzi, Honglak Lee,	<i>terpreting Neural Networks for NLP</i> , pages 217–227,	773
720	and Geoffrey Hinton. 2019. Similarity of neural	Online. Association for Computational Linguistics.	774
721	network representations revisited . In <i>Proceedings</i>		
722	<i>of the 36th International Conference on Machine</i>	Masato Mita, Taiga Someya, Ryo Yoshida, and Yohei	775
723	<i>Learning</i> .	Oseki. 2026. Language acquisition device in large	776
		language models . In <i>Proceedings of the 64th An-</i>	777
724	Eric H. Lenneberg. 1967. <i>Biological Foundations of</i>	<i>annual Meeting of the Association for Computational</i>	778
725	<i>Language</i> . Wiley, New York.	<i>Linguistics (Volume 1: Long Papers)</i> , pages 19564–	779
		19577, San Diego, California, United States. Associ-	780
726	Gaotang Li, Yuzhong Chen, and Hanghang Tong. 2025.	ation for Computational Linguistics.	781
727	Taming knowledge conflicts in language models . In		
728	<i>Proceedings of the 42nd International Conference on</i>	Masato Mita, Ryo Yoshida, and Yohei Oseki. 2025.	782
729	<i>Machine Learning</i> .	Developmentally-plausible working memory shapes	783
		a critical period for language acquisition . In <i>Proceed-</i>	784
730	Yang Liu and Chenhui Chu. 2026. Understanding the	<i>ings of the 63rd Annual Meeting of the Association</i>	785
731	prompt sensitivity . In <i>Proceedings of the 64th An-</i>	<i>for Computational Linguistics (Volume 1: Long Pa-</i>	786
732	<i>annual Meeting of the Association for Computational</i>	<i>pers)</i> , pages 9386–9399, Vienna, Austria. Associa-	787
		tion for Computational Linguistics.	788

789	Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas	Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groen-	847
790	Joseph, Nova DasSarma, Tom Henighan, Ben Mann,	evel, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling	848
791	Amanda Askell, Yuntao Bai, Anna Chen, Tom Con-	Gu, Shengyi Huang, Matt Jordan, Nathan Lambert,	849
792	erly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds,	Dustin Schwenk, Oyvind Tafjord, Taira Anderson,	850
793	Danny Hernandez, Scott Johnston, Andy Jones, Jack-	David Atkinson, Faeze Brahman, Christopher Clark,	851
794	son Kernion, Liane Lovitt, and 7 others. 2022. In-	Pradeep Dasigi, Nouha Dziri, and 24 others. 2024. 2	852
795	context learning and induction heads . <i>Transformer</i>	OLMo 2 furious . arXiv preprint arXiv:2501.00656.	853
796	<i>Circuits Thread</i> .		
797	Francesco Ortu, Zhijing Jin, Diego Doimo, Mrinmaya	Curt Tigges, Michael Hanna, Qinan Yu, and Stella Bi-	854
798	Sachan, Alberto Cazzaniga, and Bernhard Schölkopf.	derman. 2024. LLM circuit analyses are consistent	855
799	2024. Competition of mechanisms: Tracing how	across training and scale . In <i>Advances in Neural</i>	856
800	language models handle facts and counterfactuals .	<i>Information Processing Systems</i> .	857
801	In <i>Proceedings of the 62nd Annual Meeting of the</i>	Michael Tomasello. 2003. <i>Constructing a Language: A</i>	858
802	<i>Association for Computational Linguistics (Volume 1:</i>	<i>Usage-Based Theory of Language Acquisition</i> . Har-	859
803	<i>Long Papers)</i> , pages 8420–8436, Bangkok, Thailand.	vard University Press, Cambridge, MA.	860
804	Association for Computational Linguistics.		
805	Nils Reimers and Iryna Gurevych. 2017. Reporting	Yao Tong, Haonan Wang, Siquan Li, Kenji Kawaguchi,	861
806	score distributions makes a difference: Performance	and Tianyang Hu. 2026. SeedPrints: Fingerprints	862
807	study of LSTM-networks for sequence tagging . In	can even tell which seed your large language model	863
808	<i>Proceedings of the 2017 Conference on Empirical</i>	was trained from . In <i>International Conference on</i>	864
809	<i>Methods in Natural Language Processing</i> , pages 338–	<i>Learning Representations</i> .	865
810	348, Copenhagen, Denmark. Association for Compu-		
811	tational Linguistics.	Oskar van der Wal, Pietro Lesci, Max Muller-Eberstein,	866
812	Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane	Naomi Saphra, Hailey Schoelkopf, Willem Zuidema,	867
813	Suhr. 2024. Quantifying language models’ sensitiv-	and Stella Biderman. 2025. PolyPythias: Stability	868
814	ity to spurious features in prompt design or: How i	and outliers across fifty language model pre-training	869
815	learned to start worrying about prompt formatting .	runs . In <i>International Conference on Learning Rep-</i>	870
816	In <i>International Conference on Learning Representa-</i>	<i>resentations</i> .	871
817	<i>tions</i> .	George Wang, Jesse Hoogland, Stan van Wingerden,	872
818	Thibault Sellam, Steve Yadlowsky, Jason Wei, Naomi	Zach Furman, and Daniel Mufet. 2025. Differen-	873
819	Saphra, Alexander D’Amour, Tal Linzen, Jasmijn	tiation and specialization of attention heads via the	874
820	Bastings, Iulia Turc, Jacob Eisenstein, Dipanjan Das,	refined local learning coefficient . In <i>International</i>	875
821	Ian Tenney, and Ellie Pavlick. 2022. The MultiB-	<i>Conference on Learning Representations</i> .	876
822	ERTs: BERT reproductions for robustness analysis .	Alex Warstadt and Samuel R. Bowman. 2022. What	877
823	In <i>International Conference on Learning Representa-</i>	artificial neural networks can tell us about human lan-	878
824	<i>tions</i> .	guage acquisition. In <i>Algebraic Structures in Natural</i>	879
825	Aaditya K. Singh, Ted Moskovitz, Felix Hill, Stephanie	<i>Language</i> , pages 17–60. CRC Press.	880
826	C. Y. Chan, and Andrew M. Saxe. 2024. What needs	Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song	881
827	to go right for an induction head? a mechanistic study	Han, and Mike Lewis. 2024. Efficient streaming lan-	882
828	of in-context learning circuits and their formation . In	guage models with attention sinks . In <i>International</i>	883
829	<i>Proceedings of the 41st International Conference on</i>	<i>Conference on Learning Representations</i> .	884
830	<i>Machine Learning</i> .	Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and	885
831	Samuel L. Smith and Quoc V. Le. 2018. A bayesian	Yu Su. 2024. Adaptive chameleon or stubborn sloth:	886
832	perspective on generalization and stochastic gradient	Revealing the behavior of large language models in	887
833	descent . In <i>International Conference on Learning</i>	knowledge conflicts . In <i>International Conference on</i>	888
834	<i>Representations</i> .	<i>Learning Representations</i> .	889
835	Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin	Xiulin Yang, Heidi R. Getz, and Ethan Gotlieb Wilcox.	890
836	Schwenk, David Atkinson, Russell Authur, Ben	2026. Function words as statistical cues for language	891
837	Bogin, Khyathi Chandu, Jennifer Dumas, Yanai	learning . In <i>Proceedings of the 64th Annual Meet-</i>	892
838	Elazar, Valentin Hofmann, Ananya Jha, Sachin Ku-	<i>ing of the Association for Computational Linguis-</i>	893
839	mar, Li Lucy, Xinxu Lyu, Nathan Lambert, Ian Mag-	<i>tics (Volume 1: Long Papers)</i> , pages 16042–16058,	894
840	nusson, Jacob Morrison, Niklas Muennighoff, and 17	San Diego, California, United States. Association for	895
841	others. 2024. Dolma: an open corpus of three trillion	Computational Linguistics.	896
842	tokens for language model pretraining research . In	Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and	897
843	<i>Proceedings of the 62nd Annual Meeting of the As-</i>	Muhao Chen. 2023. Context-faithful prompting	898
844	<i>sociation for Computational Linguistics (Volume 1:</i>	for large language models . In <i>Findings of the As-</i>	899
845	<i>Long Papers)</i> , pages 15725–15788, Bangkok, Thai-	<i>sociation for Computational Linguistics: EMNLP</i>	900
846	land. Association for Computational Linguistics.	2023, pages 14544–14556, Singapore. Association	901
		for Computational Linguistics.	902

A Auditing the Public Suites

DataDecide 1B recipe	seed	step 0 vs. seed's step 0	step 2500 vs. seed's step 0	step 2500 vs. partner
DCLM QC 7% FW2	1	0.00	0.00	0.11
DCLM QC 7% FW3	1	0.00	0.00	0.11
Dolma 1.7 no math/code	1	0.00	0.00	0.08
Dolma 1.7 no Reddit	1	0.00	0.00	0.08
Falcon+CC QC orig. 10%	3	0.00	0.00	0.09
Falcon+CC QC Tulu 10%	3	0.00	0.00	0.09
genuine siblings (C4 vs. Dolma 1.7, same seed), step 2500				0.07–0.08

Table 4: **Six 1B runs were not initialized from their seed’s weights.** Correlation of one attention weight matrix (block 5) between checkpoints. The other 69 runs share their seed’s step-0 weights exactly.

Step-0 identity. Pythia’s standard and deduplicated models have tensor-identical step-0 weights at 70M, 160M, 410M, 1.4B and 2.8B, rounding-level identical weights at 1B (correlation 0.99993) and identical files at 6.9B and 12B. For DataDecide we compare one weight matrix of every released step-0 checkpoint with the step 0 of the same seed’s C4 run, reading the tensor directly from the hosted files.

Training start. A published step-0 checkpoint need not be the state a run started from, so we compare the earliest training checkpoint with every candidate initialization. The DataDecide 1B runs of C4, Dolma 1.7 and DCLM correlate at step 2,500 only with their own seed’s step-0 weights (0.23 vs. 0.000). PolyPythias’ “weight-seed” variants at 160M, in contrast, were trained from the standard initialization: their step-1 weights equal the standard model’s (correlation 1.0) and are unrelated to their own published step 0; we label them accordingly.

Six 1B runs with an unlisted initialization. Six 1B DataDecide runs disagree with all their nominal siblings in Figure 1a: the DCLM-QC 7% FineWeb-2/3 recipes and the Dolma 1.7 no-math-code / no-Reddit ablations of seed 1, and the Falcon-and-CC QC original / Tulu recipes of seed 3 (Table 4). Their published step-0 weights differ from their seed’s, their step-2,500 weights correlate with no seed’s step 0, and each pair’s step-2,500 weights correlate with each other as strongly as those of two genuine siblings: each pair was trained from an initialization of its own. All other 69 runs share their seed’s step-0 weights exactly, so the layout flagged exactly the six runs that differ. We exclude them from

every same-seed and different-seed comparison at 1B.

A near-duplicate Pythia pair. The released final checkpoint of pythia-2.8b-deduped is nearly identical to that of pythia-2.8b (attention-weight correlation 0.993, embedding correlation 0.985), whereas models grown from one initialization on different data retain a correlation of only 0.03–0.09 with it; we exclude the 2.8B pair.

B Head Roles

Induction: attention from the second copy of a random 128-token block to the token after its first occurrence (200 blocks). **Previous-token:** attention to position $t - 1$ on 50 natural texts (split-half reliability 0.99–1.00). **Attention sink:** attention to position 0 on the same texts. **Retrieval:** attention from the last prompt token to the counterfactual entity in the knowledge-conflict prompts. **Duplicate-token, current-token, two-back and delimiter:** attention to an earlier copy of the current token, to itself, to $t - 2$ and to punctuation, on natural text. **OV copying:** the share of positive eigenvalue mass of the head’s OV circuit composed with the embedding and unembedding (Elhage et al., 2021), computed from weights alone. An induction map is used only when the model has an induction head (maximum score > 0.3).

C The Corpus Cannot Choose the Head

Lemma 1. *Let a layer have H heads and let π permute them together with their query, key, value and output blocks. Assume (i) the initialization distribution is invariant under π ; (ii) training is π -equivariant, i.e. the update applied to $\pi\theta$ is π applied to the update of θ (true for gradient descent, Adam/AdamW, weight decay and global-norm clipping, as the loss is π -invariant); and (iii) the data and their order are independent of the initialization. Let $w(\theta)$ be the head with the largest score for a role, computed from the network’s internals, so that $w(\pi\theta) = \pi(w(\theta))$. Then for every corpus D , $P(w(\theta_T) = h \mid D) = 1/H$, and two independently initialized models trained on any corpora D, D' agree on the winning head with probability $1/H$; the expected within-layer rank correlation of their head scores is 0.*

Proof sketch. By (i)–(ii), θ_T and $\pi\theta_T$ have the same distribution given D , so $w(\theta_T)$ is uniform over heads. Independent initializations make w

and w' independent, so they agree with probability $\sum_h (1/H)^2 = 1/H$; averaging over permutations sends any rank correlation to zero in expectation. $\square$

Different-seed pairs indeed agree at -0.02 to 0.00 for all nine roles (Table 1a).

D Results at Every Size

Table 5 lists, for every DataDecide size, the recipes and seeds used, same-seed and other-seed agreement, seed identification at the end of training and in the earliest checkpoints, the corpus-distance law and the recipe’s SGD temperature. Inheritance is weak up to 20M parameters, where most models have not yet formed an induction head, and strong from 150M on.

E Controlled Pretraining

Llama-style models (RMSNorm, SwiGLU, RoPE, untied embeddings, OLMo tokenizer): 4 layers $\times$ 8 heads, $d = 256$; and 12 layers $\times$ 12 heads, $d = 768$. AdamW (learning rate 10^{-3} , $\beta = (0.9, 0.95)$, weight decay 0.1), 300 warm-up steps, cosine decay to 10%, batches of 64×512 tokens, 10,000 steps. Corpora: the first 400M tokens of single DataDecide sources for web text (C4), code (StarCoder), scientific papers (peS2o) and books (Gutenberg). Branches continue from the parent’s saved model and optimizer state at step $k \in \{100, 250, 500, 1000, 2000, 4000, 8000\}$, either switching the corpus or adding $\epsilon \cdot \text{std}(W) \cdot \mathcal{N}(0, 1)$ to every tensor (Table 6). At initialization the basin is finite: perturbations up to 1% of the weight scale leave the layout unchanged (0.91–0.99), 10% partly re-draw it (0.57), and 100% re-draw it fully (0.05–0.18).

F SGD Temperature

At fixed size we vary the batch (16, 64, 512 sequences) and the learning rate ($3 \cdot 10^{-3}$, 10^{-3} , $3 \cdot 10^{-4}$) and read the layout at step 2,000, after the critical period (Table 7). Order-only agreement is ordered by the learning rate per batch token, and tripling the learning rate acts like quartering the batch.

G No Lucky Seeds

Table 8 breaks the benchmark analysis of §4 down by task. The corpus is a significant factor on almost every task and size; the seed reaches significance

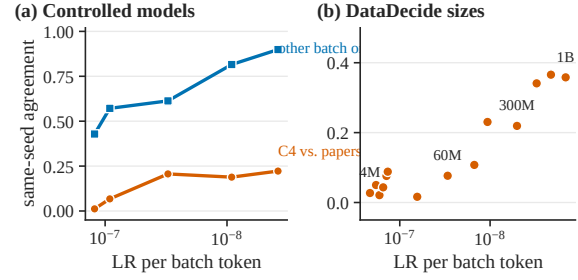


Figure 6: **Colder training, more innate.** (a) Controlled models at step 2,000: which-head agreement between runs that differ only in batch order, and between C4 and scientific-paper runs from one seed, against the learning rate per batch token. (b) DataDecide: same-seed agreement against the recipe’s learning rate per batch token.

in 0–2 of 14 sizes per task, as expected by chance, and its median variance share is zero on every task.

H The Template Habit in Detail

Prompts. Context S = the first sentence of a Para-Conflict substitution conflict; declarative: S + answer stem; QA: S + Question: q Answer: + stem; variants replace the markers (Table 3).

Items. Facts known to every compared model (the memorized answer is more likely than the counterfactual without context): 1,680 ParaConflict items over six relations, 1,299 PopQA items, and 3,533 (DataDecide) and 3,750 (OLMo 2) NQ-Swap items. Standard errors use the seed-to-seed variance pooled over four 1B recipes with three seeds each.

Sizes and public models. Table 9 gives every template at every size; Table 10 gives the trust of public base models with and without the question format.

I Artifacts, Compute and Software

Artifacts. We use the DataDecide models (Apache 2.0) and data recipes (ODC-BY), Pythia and PolyPythias (Apache 2.0), OLMo 2 (Apache 2.0), ParaConflict (Apache 2.0), NQ-Swap (MIT) and PopQA as released by their authors, all for research use consistent with their intended use; the corpora are English.

Compute. All experiments ran on NVIDIA A100 80GB GPUs, about 150 GPU-hours in total, including exploratory runs; most of it went to the con-

Parameters	Recipes	Seeds	Same seed, other corpus	Other seed, same corpus	Seed ID (%)	Seed ID at 3–9% (%)	ρ (corpus distance, agreement)	LR per batch token (10^{-9})
3.7M	18	3	0.03	0.00	56	–	–	213.6
6.0M	23	3	0.05	−0.01	77	–	–	183.1
8.5M	17	3	0.02	0.00	65	–	–	167.8
9.9M	16	3	0.04	0.00	81	72	–	152.6
14M	18	3	0.08	−0.01	83	–	–	140.4
16M	13	3	0.09	0.00	85	–	–	135.8
19M	20	3	0.02	−0.01	63	61	–	64.1
57M	17	3	0.08	0.00	98	100	−0.20 to −0.33	29.5
98M	20	3	0.11	0.01	100	–	−0.43 to −0.50	15.0
151M	15	3	0.23	−0.03	100	100	−0.47 to −0.64	10.7
320M	15	3	0.22	−0.01	100	100	−0.69 to −0.78	5.0
530M	10	3	0.34	0.01	100	–	−0.65 to −0.71	3.1
750M	10	3	0.37	0.02	100	100	−0.66 to −0.77	2.1
1.2B [†]	10	5	0.36	−0.01	100	–	−0.75 to −0.83	1.5
1.2B (final)	25	3	0.35	−0.01	100	–	−0.76 to −0.80	1.5

Table 5: **DataDecide at every size.** Which-head agreement (mean over induction, previous-token and retrieval maps; induction only where models have induction heads), leave-one-corpus-out seed identification (chance 33%; 20% with five seeds), the Spearman correlation between unigram corpus distance and same-seed agreement over corpus pairs (range over roles), and the SGD temperature of the recipe. [†]: five seeds at step 7,500 (10% of training); final: the verified 3-seed crossing at the end of training.

Step	Share	Switch to code	Noise 100%	Noise 10%	Code, 12 layers
0	0%	−0.03	0.03	0.74	−0.03
100	1%	0.09	0.36	0.94	0.40
250	2.5%	0.60	0.85	0.97	–
500	5%	0.72	0.96	0.99	–
1000	10%	0.71	0.95	0.99	0.84
2000	20%	0.81	0.98	1.00	–
4000	40%	0.86	0.96	1.00	–
8000	80%	0.95	0.98	1.00	–
exact rerun			0.97		

Table 6: **The critical period in numbers.** Which-head agreement (mean of induction and previous-token maps) between the final layout of a run intervened on at a given step and the untouched run. Step 0 for the code switch: the same seed trained on code from scratch.

Batch	LR	LR per batch token (10^{-6})	other batch order	C4 vs. papers
16	$1 \cdot 10^{-3}$	0.12	0.43	0.01
64	$3 \cdot 10^{-3}$	0.09	0.57	0.07
64	$1 \cdot 10^{-3}$	0.03	0.61	0.21
64	$3 \cdot 10^{-4}$	0.01	0.82	0.19
512	$1 \cdot 10^{-3}$	0.00	0.90	0.22

Table 7: **SGD temperature.** Same-seed which-head agreement of previous-token maps at step 2,000 between runs that differ only in batch order, and between runs on C4 and on scientific papers.

tel tests for pair-level regressions (Mantel, 1967); partial Spearman correlations by rank regression.

trolled pretraining runs, while measuring the public checkpoints takes a few minutes per model.

Software. PyTorch 2.8, Hugging Face Transformers 4.57 and Datasets 4.8, NumPy 1.26 and SciPy 1.17; models are loaded in bfloat16 with eager attention to read attention patterns. Benchmark scores are DataDecide’s released OLMES evaluations (Magnusson et al., 2025); marker counts in Dolma 1.7 come from the infini-gram index.

J Statistics

Recipe-bootstrap 95% intervals for same-seed minus other-seed agreement; unbiased two-way variance components from mean squares with F tests; permutation tests with recipe labels permuted; Man-

Benchmark	Accuracy				Correct-answer likelihood per byte			
	seed sig.	corpus sig.	seed share	corpus share	seed sig.	corpus sig.	seed share	corpus share
ARC-Easy	0/14	14/14	0.00	0.95	0/14	14/14	0.00	0.91
ARC-Challenge	1/14	10/14	0.00	0.49	0/14	14/14	0.00	0.92
BoolQ	0/14	7/14	0.00	0.25	2/14	13/14	0.00	0.55
CSQA	0/14	12/14	0.00	0.55	1/14	14/14	0.01	0.56
HellaSwag	2/14	13/14	0.00	0.60	2/14	14/14	0.00	0.96
MMLU	1/14	13/14	0.00	0.87	1/14	14/14	0.00	0.95
OpenBookQA	1/14	10/14	0.00	0.40	1/14	14/14	0.00	0.85
PIQA	0/14	14/14	0.00	0.56	1/14	14/14	0.00	0.76
Social IQa	0/14	10/14	0.00	0.30	2/14	13/14	0.00	0.48
WinoGrande	1/14	3/14	0.00	0.07	1/14	13/14	0.00	0.79
OLMES average	0/14	10/14	0.00	0.46	1/14	14/14	0.00	0.78

Table 8: **No lucky seeds.** For each benchmark, the number of the 14 model sizes at which the seed or the corpus has a significant main effect ($p < 0.05$, two-way ANOVA over recipes and seeds) and the median unbiased variance share of each factor across sizes.

Size	Question: <i>q</i> Answer:	Question: <i>q</i>	Answer:	Q: <i>q</i> A:	Query: <i>q</i> Resp.:	other entity
60M	+1.71 (0.58)	+0.73 (0.35)	+1.74 (0.59)	+0.48 (0.51)	+0.52 (0.83)	+2.14 (0.51)
90M	+1.39 (0.37)	+1.23 (0.55)	+1.51 (0.17)	+0.20 (0.44)	+0.40 (0.52)	+1.89 (0.47)
150M	+1.20 (0.57)	+1.09 (0.43)	+1.45 (0.70)	+1.16 (0.62)	+0.85 (0.59)	+1.48 (0.41)
300M	+1.52 (0.44)	+1.60 (0.60)	+1.54 (0.50)	+0.58 (0.48)	+0.19 (0.65)	+1.37 (0.41)
530M	+0.94 (0.79)	+0.96 (0.79)	+0.33 (0.70)	−0.14 (0.72)	+0.08 (0.72)	+1.51 (1.14)
750M	+1.05 (0.59)	+1.72 (0.35)	+1.35 (1.40)	+0.68 (0.50)	+0.40 (0.61)	+2.17 (0.50)
1B	+2.43 (0.79)	+2.04 (0.83)	+1.13 (0.63)	+0.39 (0.89)	+0.52 (0.83)	+3.06 (1.42)

Table 9: **The Question: habit at every size.** Flan minus no-Flan DataDecide models: added trust in the counterfactual context relative to a declarative ending, in nats (SE). The Flan template carries the effect at every size; equivalent templates do not.

Model	Instruction data in pretraining	declarative	question	gain
OLMo-1B (July 2024)	Dolma 1.7, with Flan	4.33	8.92	+4.60
Qwen2.5-1.5B	synthetic data from instruct models	−0.47	2.70	+3.18
Pythia-1B	none (the Pile)	5.47	6.89	+1.41
Pythia-1.4B	none (the Pile)	5.41	6.78	+1.38
SmolLM2-1.7B	not documented	2.89	3.93	+1.04
Gemma-2-2B	not documented	3.48	4.51	+1.03
OLMo-2-1B, before mid-training	none			−0.26 to +0.08
OLMo-2-1B, after mid-training	16.6% FLAN			+1.45 to +2.28

Table 10: **The habit in public base models.** Trust in a one-sentence counterfactual context (nats) with a declarative ending and with a Question:/Answer: ending, and the gain from the question format. The model pretrained with Flan trusts a question most.